

Dry Arthroscopic Cartilage Repair of the Knee Joint Using Umbilical Cord Mesenchymal Stem Cells: Kelly Clamp Technique



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Abstract: Among various cartilage regeneration treatments, methods using mesenchymal stem cells, whose safety and effectiveness have been verified, are emerging. Mesenchymal stem cell can be implanted through open arthrotomy or arthroscopy. Although arthroscopic surgery has the advantage of earlier recovery and less scar formation compared to open arthrotomy, dry arthroscopy is not technically easy, which is necessary for successful implantation and prevention of washout. This Technical Note will introduce an easier and more effective method of dry arthroscopic mesenchymal stem cell implantation.

Articular cartilage injury has the potential for progressive cartilage wear, which can lead to osteoarthritis and cause pain with functional disability. Because the regeneration ability of cartilage lesions is limited, various treatments are being developed.¹⁻⁴ Among various treatments, methods using mesenchymal stem cells, whose safety and effectiveness have been verified, are emerging.⁵⁻⁷

Mesenchymal stem cells can be implanted through open arthrotomy or arthroscopy. Although arthroscopic surgery has the advantage of earlier recovery and less scar formation compared to open arthrotomy, dry arthroscopy is not technically easy, which is necessary for successful implantation and prevention of washout.

Previous studies have proposed various methods of creating a working space for dry arthroscopy. This Technical Note will introduce an easier and more

effective method of dry arthroscopic mesenchymal stem cell implantation.

Surgical Technique (with Video Illustration)

Patient Positioning and Arthroscopic Examination

The patient underwent standard knee arthroscopy under spinal anesthesia in the supine position. A thorough arthroscopic examination was performed using anteromedial (AM) and anterolateral (AL) portals. The cartilage defect was debrided with a curette until the subchondral bone was exposed. After debridement, multiple holes were created within the clear margin of cartilage defect. Using a drill bit, we first made multiple 4 mm diameter main holes, then drilled additional multiple 2 mm diameter holes between the main holes (Fig 1).

Preparation

In January 2012, the Korea Food and Drug Administration approved Cartistem (Medipost, Seongnam, Gyeonggi-do, South Korea) as a treatment for cartilage defect. The therapeutic dose of Cartistem was determined after confirming the size of the cartilage defect using magnetic resonance imaging. Mesenchymal stem cells were mixed with 4% hyaluronic acid before implantation according to the manufacturer's instructions.

Dry Arthroscopy With a Kelly Clamp Technique and Implantation of Mesenchymal Stem Cells

Preparation of the defect site was confirmed, and an additional portal was made in front of the medial collateral ligament and a Kelly clamp was inserted

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The authors report that they have no conflicts of interest in the authorship and publication of this article. Full ICMJE author disclosure forms are available for this article online, as [supplementary material](#).

Received February 8, 2023; accepted April 11, 2023.

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2212-6287/23201

<https://doi.org/10.1016/j.eats.2023.04.004>



Fig 1. Arthroscopic view of the right knee through AL portal, while using the AM portal as the working portal. (A) The margin of a full-thickness cartilage defect (*star*) in the MFC was debrided. (B) Multiple drill holes. AL, anterolateral; AM, anteromedial; MFC, medial femoral condyle.

(Fig 2). After the soft tissue was retracted to create sufficient working space, fluids were drained from the knee joint space to prepare implantation of the Cartistem under dry arthroscopy. Additionally, a spinal needle with a suction drain attached was introduced to minimize visual impairment due to bleeding (Fig 3).

After that, Cartistem was first implanted in the drilled holes, and residual amount was applied to the surface of the debrided subchondral bone (Fig 4). To minimize washout, the Cartistem was first implanted from the bottom of the defect site during knee flexion and gradually to the top along with knee extension. After implantation, the range of motion was carefully checked to ensure that the Cartistem was in place. The wound was closed, and a cylinder splint was applied.

The pearls and pitfalls of this surgical technique are described in Table 1, while the tips are described in Table 2. A step-by-step technique summary is presented in Table 3.

Rehabilitation

A cylinder splint was applied immediately after surgery and was replaced by a knee brace on the first day after surgery. With the brace applied, the patient began

quadriceps exercise and straight leg raise exercise. On postoperative day 2, continuous passive motion was initiated within the tolerable range. The angle of the brace was gradually increased, and non-weightbearing gait using crutches was maintained for 6 weeks after surgery.

Discussion

Recently, the treatment of cartilage injury using mesenchymal stem cells has been on the rise.⁷⁻⁹ Among them, Cartistem is mesenchymal stem cells derived from allogenic human umbilical cord blood. Although mesenchymal stem cells can be derived from bone marrow and adipose tissue, umbilical cord contains more stem cells. Successful outcome of allogenic human umbilical cord blood-derived mesenchymal stem cell implantation has been reported in several previous studies.¹⁰⁻¹³

The procedure for Cartistem can be performed through open arthrotomy or arthroscopic surgery. Compared with arthrotomy, arthroscopic surgery has the advantages of better visualization, early rehabilitation, and less scar formation. However, arthroscopic

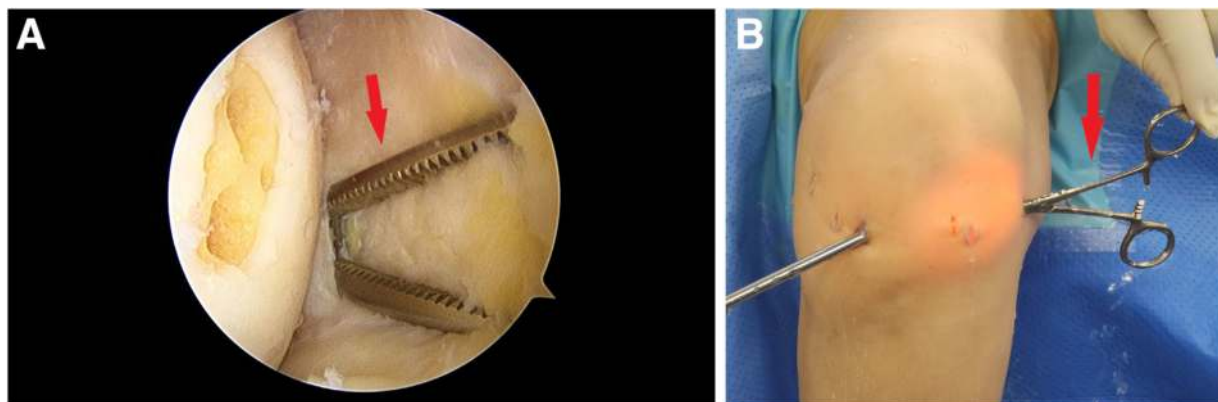


Fig 2. Arthroscopic view of the right knee through AL portal, while a Kelly clamp was introduced through far medial portal. (A) A Kelly clamp was inserted through far medial portal (*arrow*). (B) Extra-articular view of the Kelly clamp through the far medial portal (*arrow*). AL, anterolateral; AM, anteromedial.

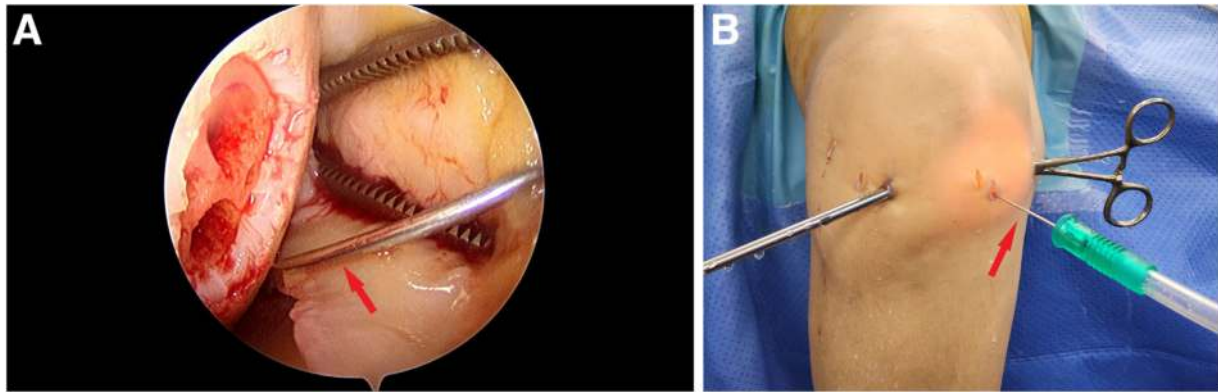


Fig 3. Arthroscopic view of the right knee through AL portal, while a spinal needle with a suction drain attached was introduced. (A) A spinal needle was inserted (arrow). (B) Extra-articular view of the spinal needle with a suction drain attached (arrow). AL, anterolateral.

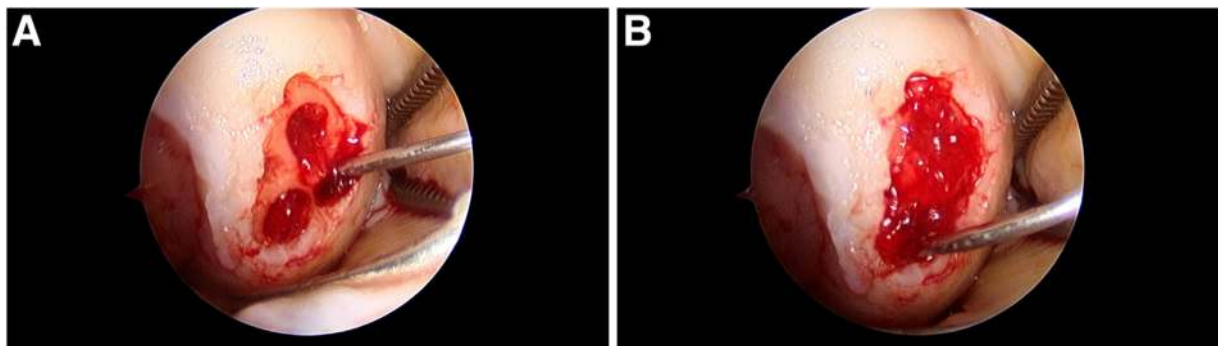


Fig 4. Arthroscopic view of the right knee through AL portal, while a Kelly clamp was introduced through far medial portal. (A) Cartistem was first implanted in the drilled holes. (B) Residual amount was applied to the surface of the debrided subchondral bone surface. AL, anterolateral.

Table 1. Pearls and Pitfalls of Dry Arthroscopic Cartilage Repair Using Cartistem

Pearls

- Appropriate planning of the portal location is essential.
- Complete exposure of the cartilage lesion is necessary and instrumental access to the entire drilled hole is required.
- Drilling should be performed with care so that adjacent holes do not connect and cause subchondral fractures.

Pitfalls

- After the soft tissue was retracted by Kelly clamp, fluids were drained from the knee joint space to prepare implantation of the Cartistem under dry arthroscopy.
- During the implantation of Cartistem, the surgeon should try to avoid excessive handling and manipulation to maximize cell viability.

surgery is not easy because it is difficult to secure enough working space because of soft tissue and dilution of mesenchymal stem cells by fluids. For this reason, open arthrotomy has been widely used so far.¹⁴

Dry arthroscopy was introduced to compensate the difficulties of arthroscopic surgery, and several

techniques have been developed for this purpose.¹⁵⁻²⁰ Conventional dry arthroscopy was performed by simply draining joint fluids, but it was not easy because of soft tissue collapse and insufficient working space. Heng et al.¹⁹ used a switching stick or probe as a soft tissue retractor and a traction suture to keep the soft tissue away from the lesion. However, the use of a switching stick and probe can only retract a small amount of tissue. Siebold et al.¹⁵ developed a technique for placing the patient in a prone position. They placed a suture through the lateral patellar retina with a weight attached. However, reversal of the anatomical situation can be difficult even for experienced arthroscopic surgeons. It also limits the ability to perform diagnostic arthroscopy of the entire knee. DeFroda et al.¹⁷ used CO₂ gas to keep the knee joint dry. Dry arthroscopy with CO₂ gas is convenient but expensive and has the potential for subcutaneous emphysema and air embolism. Sadlik et al.¹⁶ created a working space by inserting a retraction plate. However, this technique cannot be performed without this special device. Oh et al.¹⁸ recommended simple retraction using crossed

Table 2. Tips for Dry arthroscopic Cartilage Repair Using Cartistem

During dry arthroscopy, minor fogging of the lens can be addressed by touching the camera to fatty tissues.

Use a spinal needle with a suction drain attached to minimize visual impairment due to bleeding.

Cartistem was implanted from bottom of the defect site during knee flexion and gradually to the top along with knee extension.

Table 3. Step-by-Step Technique of Dry arthroscopic Cartilage Repair Using Cartistem

1. Position the patient as for standard knee arthroscopy and administer spinal anesthetic.
2. Perform diagnostic arthroscopy and identify any coexisting conditions that require treatment through standard anterolateral and anteromedial portals.
3. Visualize cartilage lesions in their entirety to ensure adequate access for arthroscopic treatment.
4. The cartilage defect was debrided with a curette until the subchondral bone was exposed.
5. After debridement, multiple holes were created within the clear margin of cartilage defect.
6. Preparation of the defect site was confirmed, and an additional portal was made in front of the medial collateral ligament and a Kelly clamp was inserted.
7. After the soft tissue was retracted to create sufficient working space, fluids were drained from the knee joint space to prepare implantation of the Cartistem under dry arthroscopy.
8. Additionally, a spinal needle with a suction drain attached was introduced to minimize visual impairment due to bleeding.
9. Mesenchymal stem cells (Cartistem) were mixed with 4% hyaluronic acid prior to implantation according to the manufacturer's instructions.
10. Cartistem was first implanted in the drilled holes, and residual amount was applied to the surface of the debrided subchondral bone. In order to minimize washout, the Cartistem was first implanted from the bottom of the defect site during knee flexion and gradually to the top along with knee extension.
11. After implantation, the range of motion was carefully checked to ensure that the Cartistem was in place.
12. The wound was closed, and cylinder splint was applied.

spinal needles and a suture receiver. This technique does not require any special equipment, but it is not convenient.

In this study, dry arthroscopy using a simple instrument, a Kelly clamp, was described. Our technique is a simpler and more effective procedure for mesenchymal stem cell implantation. Additionally, this technique can be applied to any compartment of the knee joint, depending on whether the Kelly clamp is inserted medially or laterally.

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